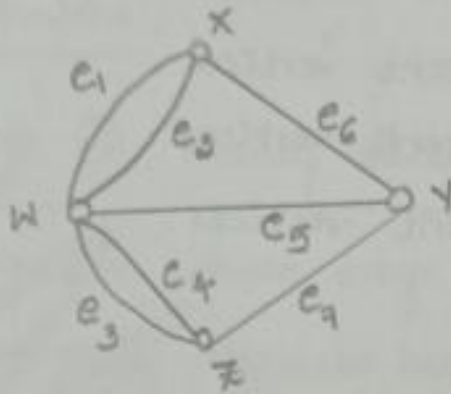


Graph Theory

A graph G is an ordered pair (V, E) consisting of :

A vertex set $V = \{W, X, Y, Z\}$

An edge set $E = \{e_1, e_2, e_3, e_4, e_5, e_6, e_7\}$



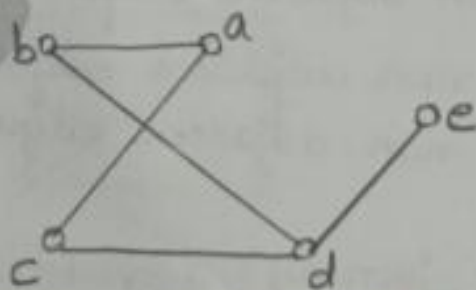
A graph G is an ordered triple (V, E, f) consisting of

- V is a set of nodes, points or vertices.

- E is a set whose elements are known as edges or lines.

- f is a function that maps each element of E to an unordered pair of vertices in V .

Adjacent vertices



Edge (a, b) joins vertices a & b

vertex ' a ' is adjacent to vertex ' b '.

or a & b are adjacent vertices.

- iii⁴
- a & c are adjacent vertices.
 - b & d are adjacent vertices.
 - d & e are adjacent vertices.

Non adjacent vertices

- a & d are non adjacent vertices.
- a & e are non adjacent vertices.
- b & e are non adjacent vertices.
- c & e are non adjacent vertices.

Adjacent edges:

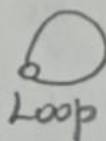
- (a, b) & (b, d) are adjacent edges.
- (a, b) & (a, c) are adjacent edges.
- (b, d) & (c, d) are adjacent edges.
- (b, d) & (d, e) are adjacent edges.
- (a, c) & (c, d) are adjacent edges.
- (c, d) & (d, e) are adjacent edges.

Non-adjacent edges:

- (a, b) & (c, d) are non adjacent edges.
- (a, c) & (b, d) are non adjacent edges.
- (a, c) & (d, e) are non adjacent edges.

Loop: It is an edge joining a vertex to itself.

$$\forall v \in V \quad (v, v) \in E$$



Multiple edges: Edges have the same pair of endpoints



Multiple edges.

Types of Graphs :

1. Null graph (or) Totally disconnected graph :

A graph whose edge set is empty.

→ It has only vertices but no edges.

a. •

b. • •

c. • • •

• • •

2. Trivial graph :

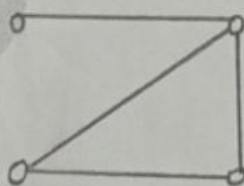
A graph having only one vertex (But no edges)

•
v

→ It has only one vertex & no edges.

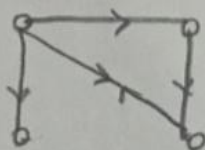
3. Undirected graph :

A graph in which all edges are undirected.



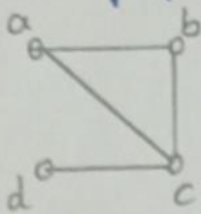
4. Directed graph (Digraph) :

A graph in which all the edges are directed.



5. Finite graph:

A graph having finite no of vertices & edges.

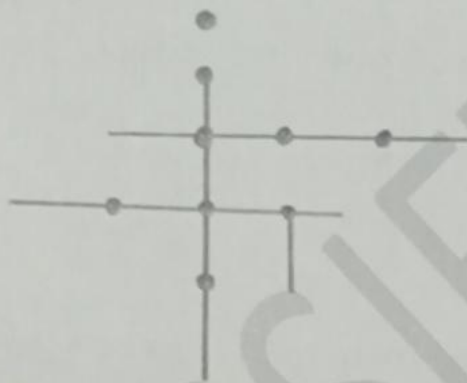


$$V = \{a, b, c, d\} = \text{finite}$$

$$E = \{ab, ac, bc, cd\}$$

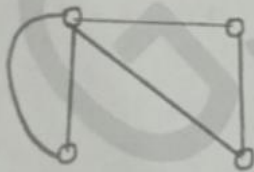
6. Infinite graph:

A graph having infinite no of vertices & infinite no of edges.



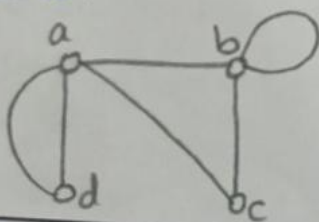
Multigraph:

A graph having no loops but having multiple edges.



Pseudo graph:

A graph which allows both loops & multiple edges in it.



Simple graph

A graph having no loops & no multiple edges in it.



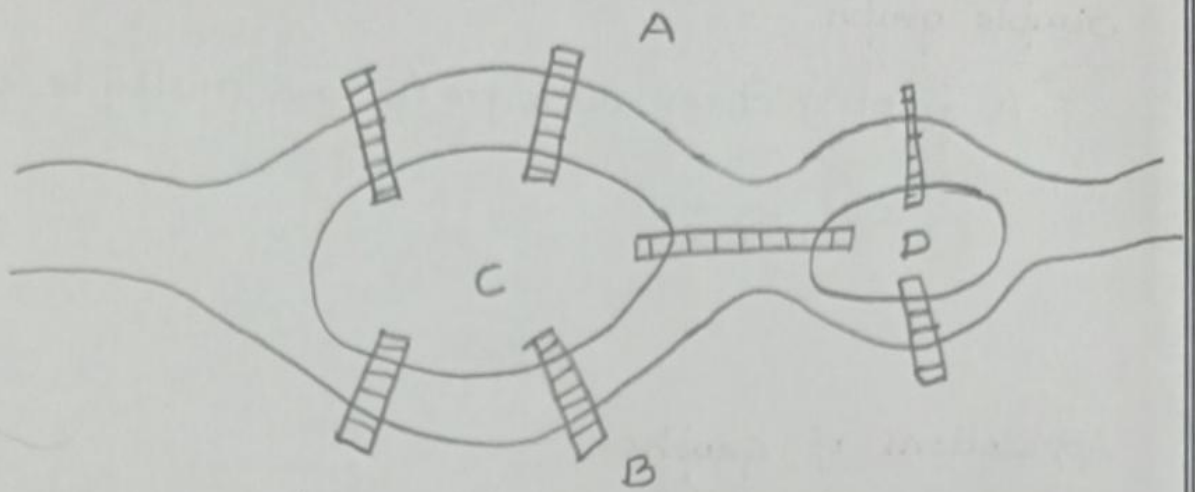
Applications of Graphs

Konigsberg Bridge Problem:

This is the best example in graph theory. It was a long-standing problem until solved by Leonhard Euler in 1736, Euler wrote the first paper ever in graph theory & thus became the originator of the theory of graphs as well as of the rest of topology. The problem is written below.

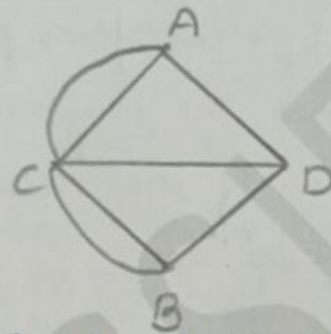
Two islands C & D, formed by the Pregel River in Konigsberg were connected to each other & to the banks A & B with seven bridges, as shown in fig (a). The problem was to start at any of the four land areas of the city A, B, C & D, walk over each of the seven bridges exactly once, & return to the starting point (without swimming across the river, of course).

Euler represented this situation by means of a graph, as shown in fig (b). The vertices represent the land areas & the edges represent the bridges.



(a) Königsberg bridge problem

Euler proved that a solution for this problem does not exist.



(b) Graph of Königsberg bridge problem.

* Utilities problem: There are three houses (Fig(a)) H_1, H_2 & H_3 each to be connected to each of the three utilities - Water (W), Gas (G) & Electricity (E) - by means of channel. Is it possible to make such connections without any crossovers of the channels.

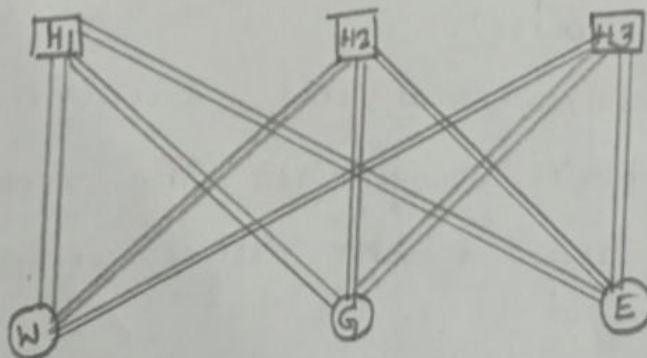
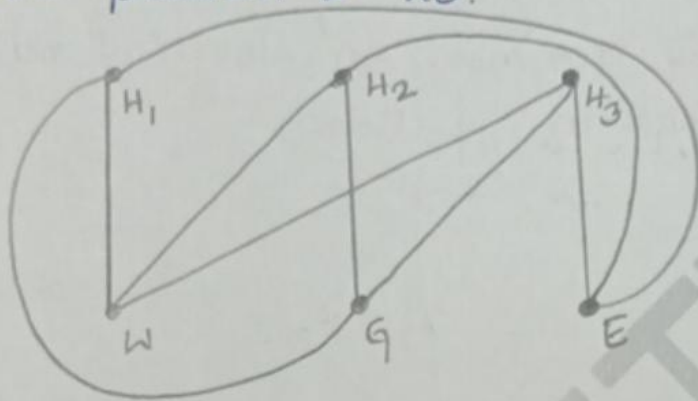


Fig (a). Three utilities problem

Fig (b) shows how this problem can be represented by a graph - the channels are shown as edges while the houses & utility supply centers are vertices.

The graph in fig (b) cannot be drawn in the plane without edges crossing over. Thus the answer to the problem is no.



Fig(b). Graph of three utilities problem

*Electrical Network problems: Properties (such as transfer function & input impedance) of an electrical network are functions of only two factors:

1. The nature & value of the elements forming the network, such as resistors, inductors, transistors & so forth.
2. The way these elements are connected together, that is, the topology of the network.

Since there are only a few different types of electrical elements, the variations in networks are chiefly due to the variations in topology. Thus electrical networks analysis & synthesis are mainly the study of network topology. In the topological study of electrical networks, factor 2 is separated from 1 & is

studied independently.

The topology of a network is studied by means of its graph. In drawing a graph of an electrical network the junctions are represented by vertices, & branches (which consist of electrical elements) are represented by edges, regardless of the nature & size of the electrical elements. An electrical network & its graph are shown in fig. below.

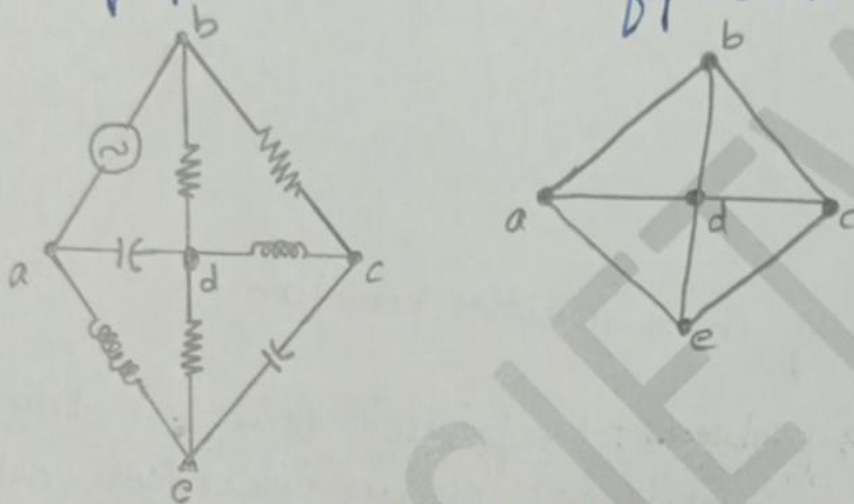
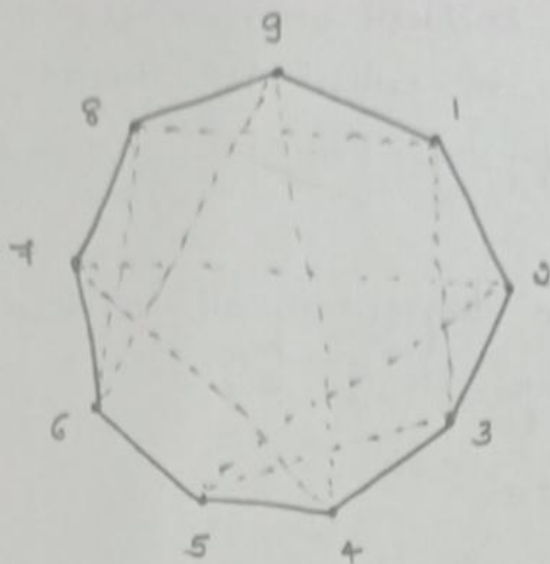


Fig. Electrical network & its graph.

Seating problem: Nine members of a new club meet each day for lunch at a round table. They decide to sit such that every member has different neighbors at each lunch. How many days can this arrangement last?

This situation can be represented by a graph with nine vertices $\Rightarrow$ each vertex represents a member, & an edge joining 2 vertices represents the relationship of sitting next to each other.



Arrangements at a dinner table.

Seating arrangements - these are 1234567891 & 1358749681. It can be shown by graph-theoretic considerations that there are only two more arrangements possible. They are 1573928461 & 1795836241. In general it can be shown that for n people the no of such possible arrangements is

$$\frac{n-1}{2}, \text{ if } n \text{ is odd}$$

$$\frac{n-2}{2}, \text{ if } n \text{ is even.}$$

Incidence & Degree:

When a vertex v_i is an end vertex of some edge e_j , v_i & e_j are said to be incident with each other. In fig. for example, edges e_2, e_6 & e_7 are incident with vertex v_4 .

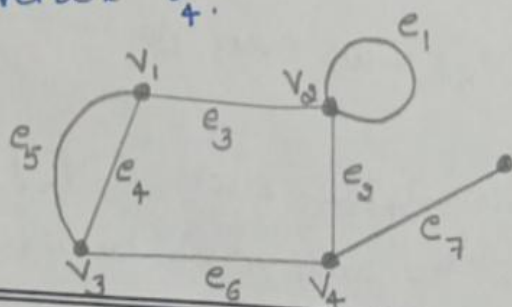


Fig. A graph with 5 vertices & 7 edges.

The no of edges incident on a vertex v_i , with self-loops counted twice, is called the degree, $d(v_i)$, of vertex v_i . In above fig, for example, $d(v_1) = d(v_2) = d(v_4) = 3$, $d(v_3) = 4$ & $d(v_5) = 1$.

The sum of the degrees of all vertices in G is twice the no of edges in G . i.e.,

$$\sum_{i=1}^n d(v_i) = 2e.$$

Theorem

The no of vertices of odd degree in a graph is always even.

Proof :- If we consider the vertices with odd & even degrees separately, the quantity in the left side of eqⁿ $\sum_{i=1}^n d(v_i) = 2e$ can be expressed as the sum of two sums, each taken over vertices of even & odd degrees, respectively, as follows:

$$\sum_{i=1}^n d(v_i) = \sum_{\text{even}} d(v_j) + \sum_{\text{odd}} d(v_k). \quad \text{--- (1)}$$

Since the LHS in eqⁿ (1) is even, & the first expression on the RHS is even (being a sum of even no's), the second expression must also be even:

$$\sum_{\text{odd}} d(v_k) = \text{an even no} \quad \text{--- (2)}$$

Because in eqⁿ (2) each $d(v_k)$ is odd, the total no of terms in the sum must be even to make the sum an even no.

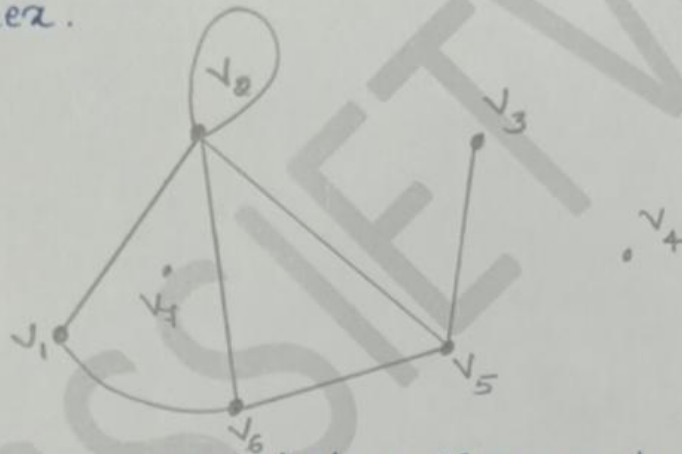
Hence the theorem.

Isolated vertex ;

A vertex having no incident edge is called an isolated vertex. In other words, isolated vertices are vertices with zero degree. vertices v_4 & v_7 in below fig. are isolated vertices.

Pendant vertex:

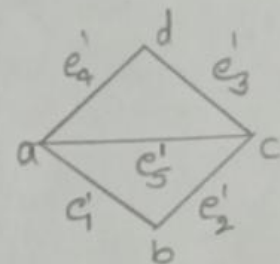
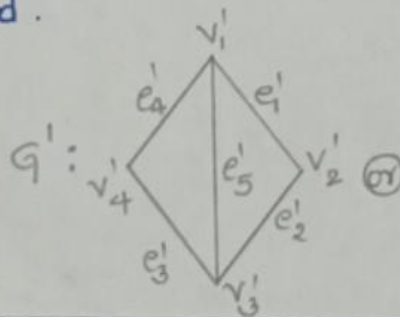
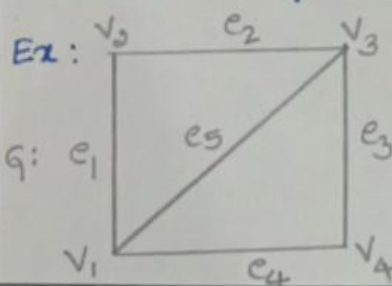
A vertex of degree one is called a pendant vertex (or) an end vertex v_3 in below fig. is a pendant vertex.



Graph containing isolated vertices, pendant vertex.

Isomorphic graph:

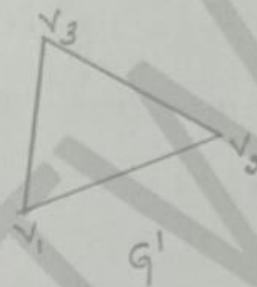
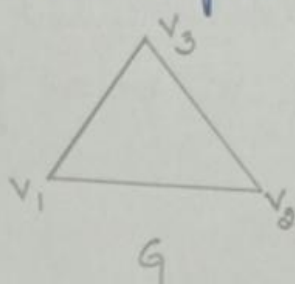
Two graph $G(V, E)$ & $G'(V', E')$ are said to be isomorphic if there exist one-to-one correspondence b/w vertices & also b/w edges $\ni$ incidence relation is preserved.



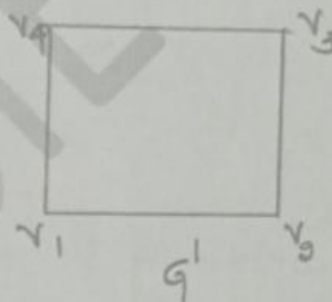
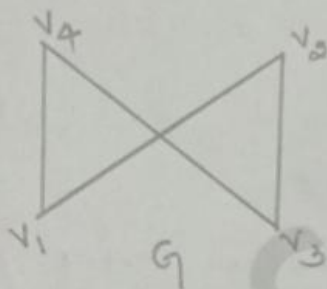
$\therefore G \cong G' \therefore G \& G'$ are isomorphic.

Note: Any two graphs are said to be isomorphic if they satisfy the following conditions

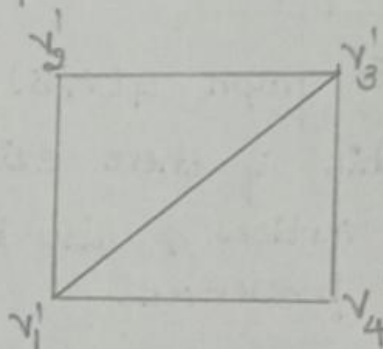
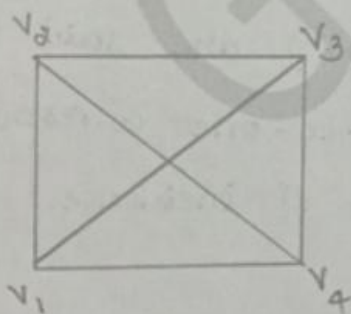
- 1) Both the graphs must have equal no of vertices.
- 2) Both the graphs must have equal no of edges.
- 3) Both the graphs must have equal no of vertices with same degree.



$$\therefore G \cong G'$$



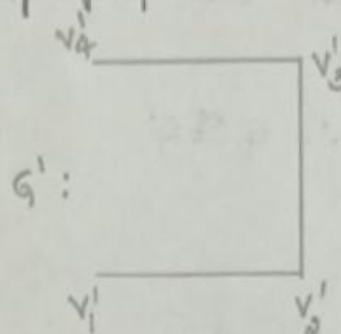
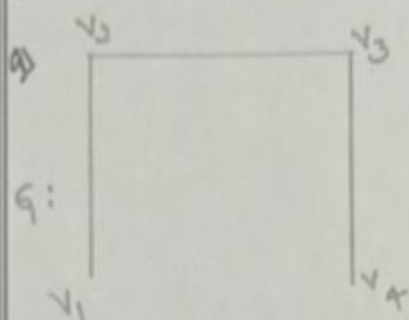
$$\therefore G \cong G'$$



$$\therefore G \not\cong G'$$

Problems

1) Show that the following graphs are isomorphic.



$$|V| = 4$$

$$|V'| = 4$$

$$|E| = 3$$

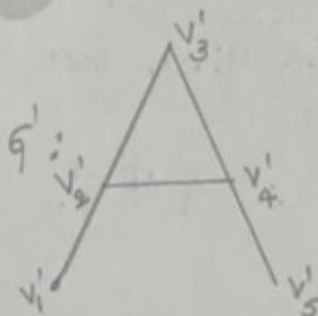
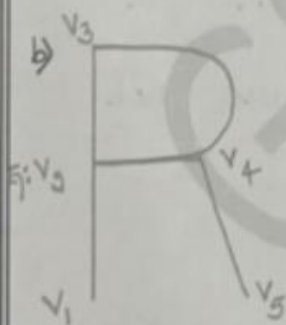
$$|E'| = 3$$

$$\langle v_1, v_2, v_3, v_4 \rangle \Leftrightarrow \langle v'_1, v'_2, v'_3, v'_4 \rangle$$

Here there is one-one correspondence b/w the vertices & also the edges & incidence relation is preserved.

$\therefore$ The given graph is isomorphic.

$$\text{ie., } G \cong G'.$$



$$|V| = 5$$

$$|V'| = 5$$

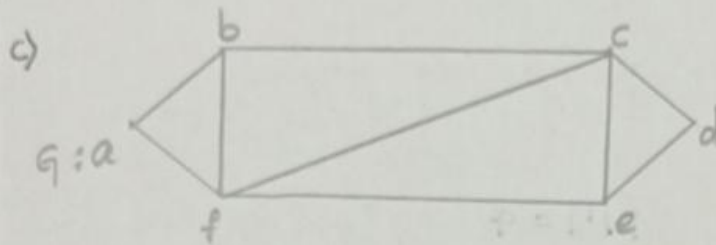
$$|E| = 5$$

$$|E'| = 5$$

$$\langle v_1, v_2, v_3, v_4, v_5 \rangle \Leftrightarrow \langle v'_1, v'_2, v'_3, v'_4, v'_5 \rangle$$

Here there is one-one correspondence b/w the vertices & also the edges & incidence relation is preserved.

$$\therefore G \cong G'$$



$$|V| = 6$$

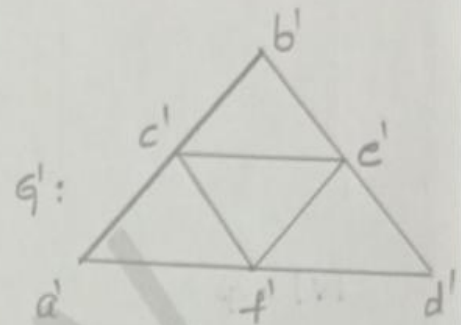
$$|E| = 9$$

Since in the given graph G has 2 vertices of degree 4 i.e., (f & c) where as the graph G' has 3 vertices of degree 4.

$\therefore$ They cannot be one-to-one correspondence b/w vertices & edges of the 2 graph.

$\therefore$ The given two graphs are not isomorphic.

$$\therefore G \not\cong G'$$



$$|V'| = 6$$

$$|E'| = 9$$

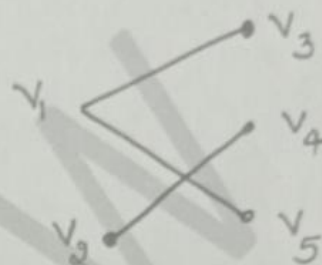
Bi-Partite graph:

A graph G is said to be bi-partite if the vertex set V is partitioned into two non-empty sets i.e., U & W $\exists$ every edge has one end vertex in U & other end vertex in W .

Ex: $V = \{v_1, v_2, v_3, v_4, v_5\}$

$$U = \{v_1, v_2\}$$

$$W = \{v_3, v_4, v_5\}$$



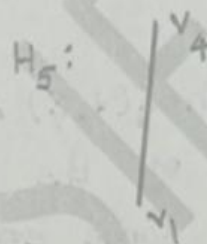
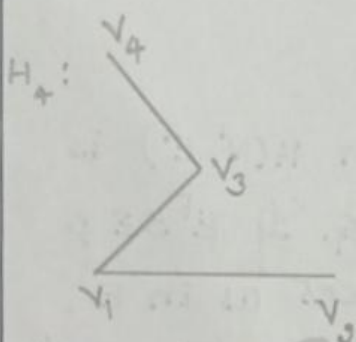
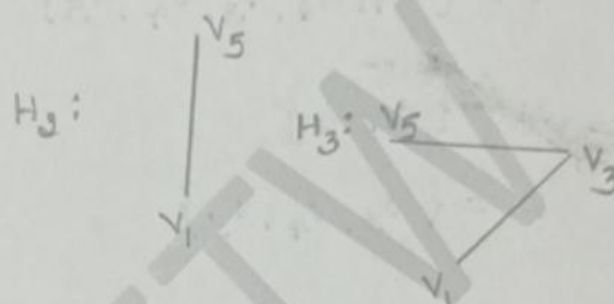
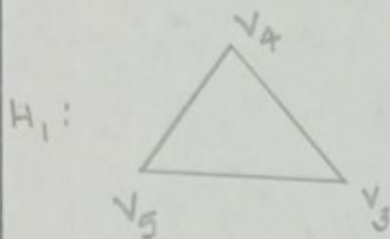
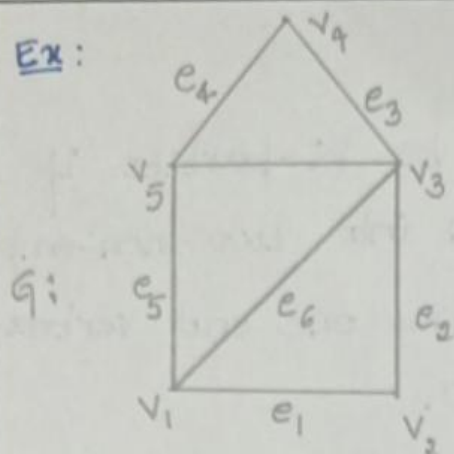
Sub graphs:

Consider a graph $G(V, E)$ a graph $H(V', E')$ is said to be subgraph of the graph G . If $E' \subseteq E$ & $V' \subseteq V$ $\exists$ E' have the same end vertices as in E .

The symbol from set theory $H \subseteq G$, is used in stating "H is a subgraph of G".

- * Every graph is its own subgraph.
- * A subgraph of a subgraph of G is a subgraph of G .
- * A single vertex in a graph G is a subgraph of G .
- * A single edge in G , together with its end vertices, is also a subgraph of G .

Ex:



Here we observe that H_1, H_2, H_3, H_4 are subgraph of G but H_5 is not a subgraph of G . Since v_4 is not incident with v_1 .

Edge disjoint:

Two graph H_1 & H_2 of a graph G are said to be edge disjoint if they donot have any edge in common.

Vertex disjoint: Two graphs H_1 & H_2 of a graph G are said to be vertex disjoint if they donot have any vertex in common.

Walk:

A walk of a graph G is a finite alternating sequence of vertices & edges which begins with a vertex & ends with a vertex.

* A walk is said to be an open walk if initial & end vertices are different.

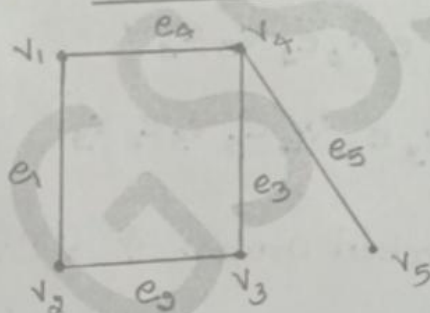
* A walk is said to be close walk if initial & end vertices are same.

Note: In a walk both vertices & edges can be repeated.

The no of edges in a walk is called length of a walk.

If length of a walk is zero then it's called Trivial walk.

Ex:



$v_1 e_1 v_2 e_2 v_3 e_3 v_4 e_5 v_5 \rightarrow$

walk of length 4.
{open walk}

$v_1 e_4 v_4 e_3 v_3 e_2 v_2 e_1 v_1 \rightarrow$

{close walk} length 4.

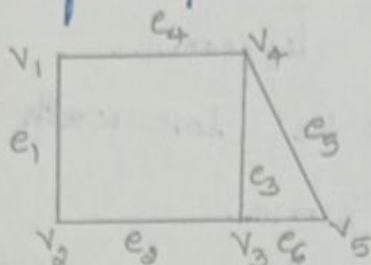
Ex: Write an walk from v_2 to v_5

$$\left. \begin{array}{l} v_2 e_2 v_3 e_3 v_4 e_5 v_5 \\ v_2 e_1 v_1 e_4 v_4 e_5 v_5 \end{array} \right\} \text{Length 3 open walk.}$$

Path:

An open walk without the repetition of vertices is called a path.

The no of edges in a path is called the length of a path.

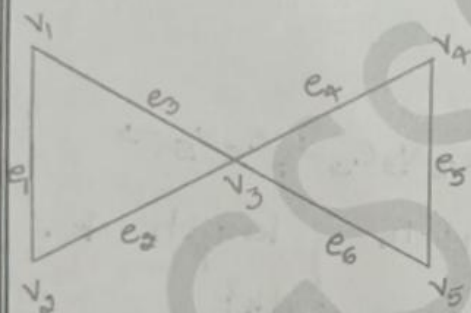


$$\cdot v_1 e_1 v_2 e_2 v_3 e_3 v_4 e_5 v_5$$

$$\cdot v_1 e_4 v_4 e_5 v_5$$

Circuit:

A circuit of graph G is a closed walk in which vertices can be repeated but edges cannot be repeated.



$$\cdot v_1 e_1 v_2 e_2 v_3 e_4 v_4 e_5 v_5 e_6 v_3 e_3 v_1$$

$$\cdot v_1 e_1 v_3 e_3 v_1$$

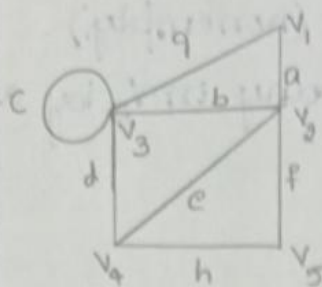
$$\cdot v_3 e_4 v_4 e_5 v_5 e_6 v_3$$

Connected graphs, Disconnected graphs & Components.

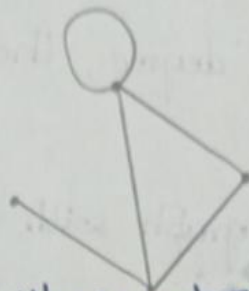
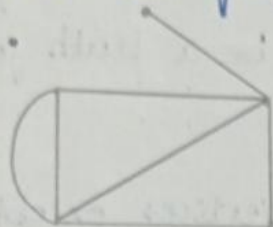
A graph G is said to be a connected if there is atleast one path b/w every pair of vertices in G . otherwise, G is disconnected.

A null graph of more than one vertex is disconnected.

Connected subgraphs is called a component.



Connected graph



A disconnected graph with 2 components.

Theorem 1:

A graph G is disconnected iff its vertex set V can be partitioned into 2 nonempty, disjoint subsets V_1 & V_2 $\exists$ there exists no edge in G whose one end vertex is in subset V_1 & the other in subset V_2 .

Proof: Suppose that such a partitioning exists.

Consider two arbitrary vertices a & b of G , $\exists$ $a \in V_1$ & $b \in V_2$. No path can exist b/w vertices a & b ; otherwise, there would be at least one edge whose one end vertex would be in V_1 & the other in V_2 . Hence, if a partition exists, G is not connected.

Conversely, let G be a disconnected graph. Consider a vertex a in G . Let V_1 be the set of all vertices that are joined by paths to a . Since G is disconnected, V_1 does not include all vertices of G .

The remaining vertices will form a (nonempty) set V_2 . No vertex in V_1 is joined to any in V_2 by an edge. Hence the partition.

Theorem 2:

If a graph (connected or disconnected) has exactly two vertices of odd degree, there must be a path joining these two vertices.

Proof: Let G be a graph with all even vertices except vertices v_1 & v_2 , which are odd. From Theorem 1, which holds for every graph & therefore for every components of a disconnected graph, no graph can have an odd no of odd vertices.

$\therefore$, in graph G , v_1 & v_2 must belong to the same component, & hence must have a path b/w them.

Theorem 3:

A simple graph (ie., a graph without parallel edges (or) self loops) with n vertices & k components can have at most $(n-k)(n-k+1)/2$ edges.

Proof: Let the no of vertices in each of the k components of a graph G be $n_1, n_2, \dots, n_k$. Thus we have

$$n_1 + n_2 + \dots + n_k = n,$$

$$n_i \geq 1.$$

The proof of the theorem depends on an algebraic inequality

$$\sum_{i=1}^k n_i^2 \leq n^2 - (k-1)(2n-k)$$

Now the maximum no of edges in the i^{th} component of G (which is a simple connected graph) is $\frac{1}{2} n_i (n_i - 1)$.

$\therefore$ The maximum no of edges in G is

$$\frac{1}{2} \sum_{i=1}^k (n_i - 1) n_i = \frac{1}{2} \left(\sum_{i=1}^k n_i^2 \right) - \frac{n}{2}$$

$$\leq \frac{1}{2} [n^2 - (k-1)(2n-k)] - \frac{n}{2},$$

$$= \frac{1}{2} \cdot (n-k)(n-k+1).$$